Warming of the Sea Floor

Determining responses to warming of Otago region benthic communities using heated panels: Design experiment of an in-situ manipulation

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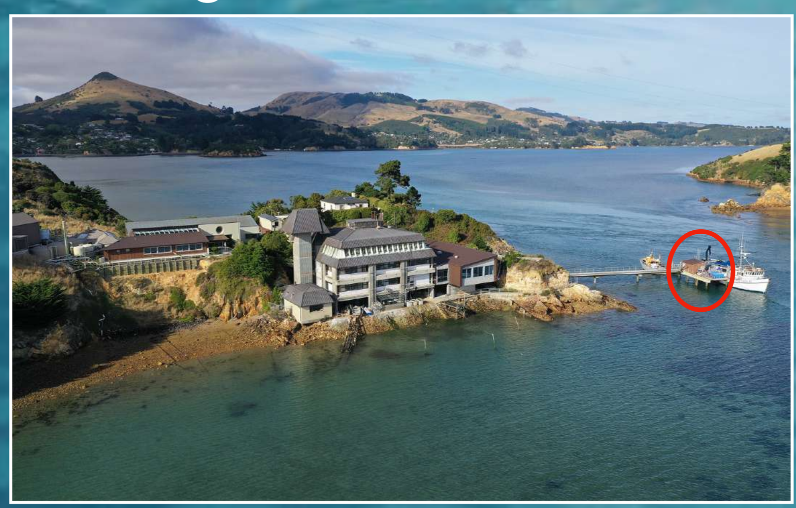
Coastal Oceans around New Zealand are forecast to warm: How will we predict warming influence on benthic communities through changes in recruitment, growth and community composition?

OTAGO REGION EXPERIMENT

Coastal seas of New Zealand are predicted to be +1°C and +2°C warmer over the next 50 and 100 years. Bespoke heated settlement panels technology will allow us to study responses to climate change in the Otago Peninsula benthic ecosystem. Sponges, Ascidians, Bryozoa and Polychaete tube-worms taxa will be analysed as an example of potential changes. These mesocosm experiments will provide an indicator of how larger scale changes in community structure may be driven by species-specific responses to warming.

Ascidians / Bryozoas / Polychaetes / Sponges



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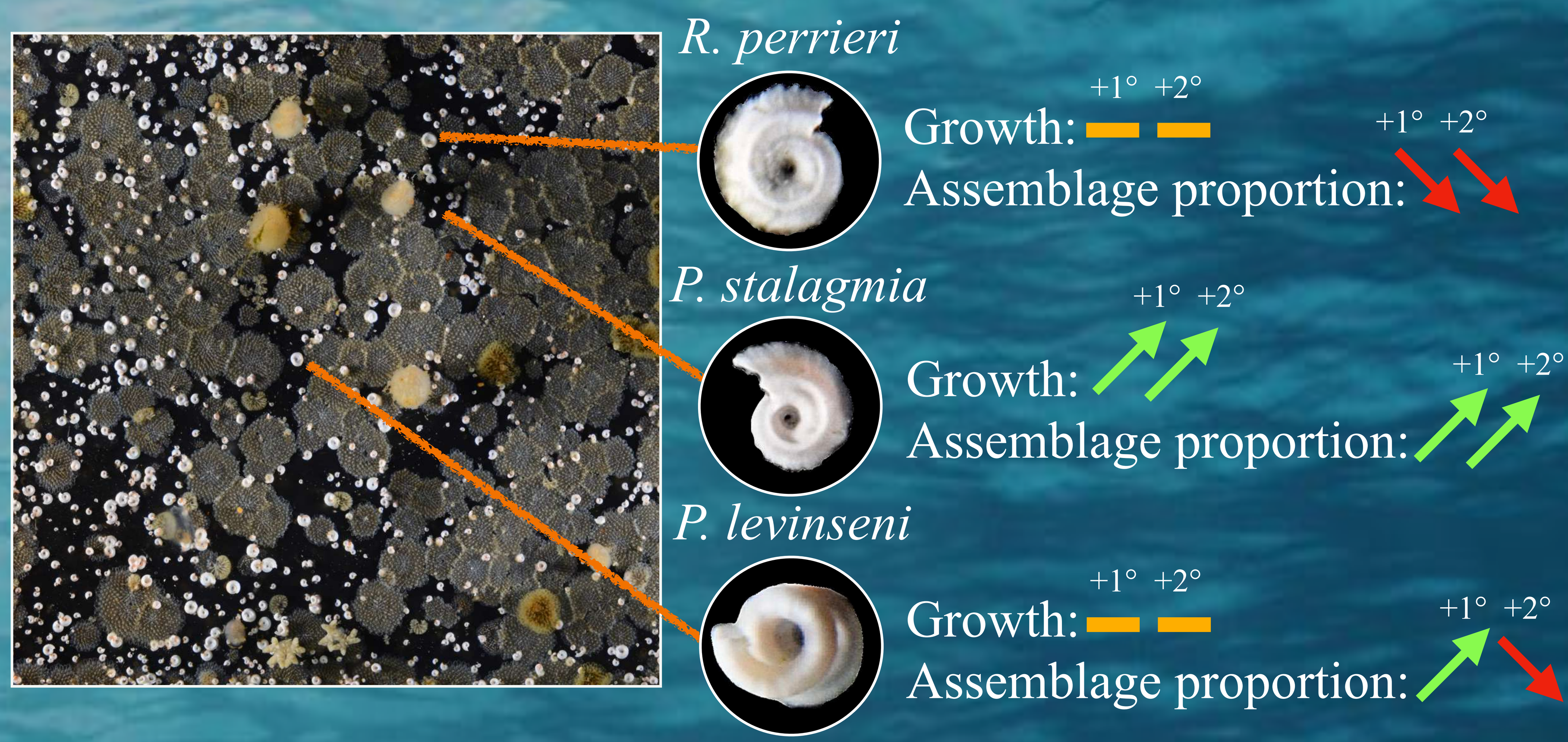
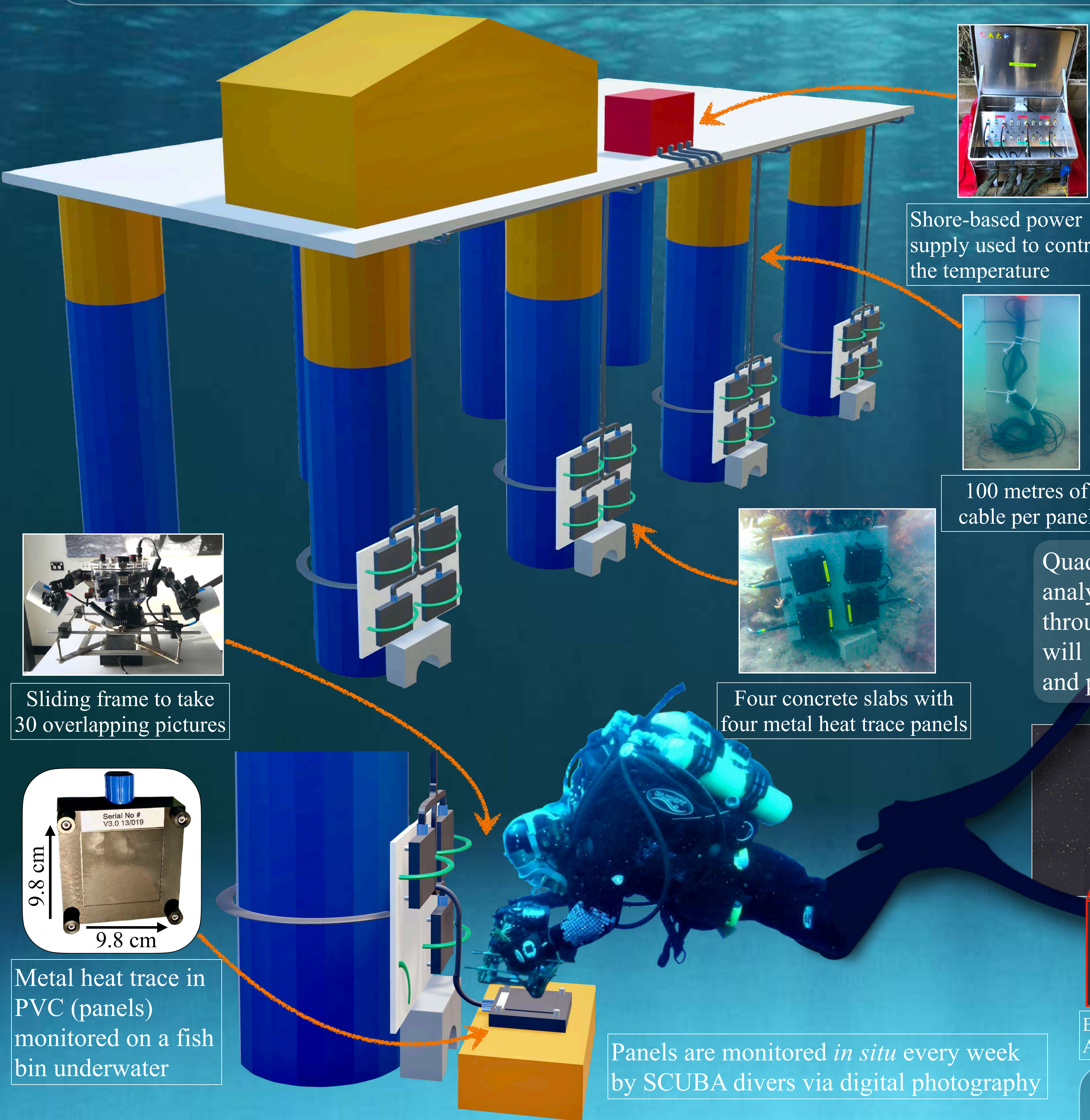
EXAMPLE OF RESULTS

Warming drives significant changes to benthic communities in the Antarctic Peninsula (tube-worms as model organisms).

Responses of tube worms at +1°C and +2°C:



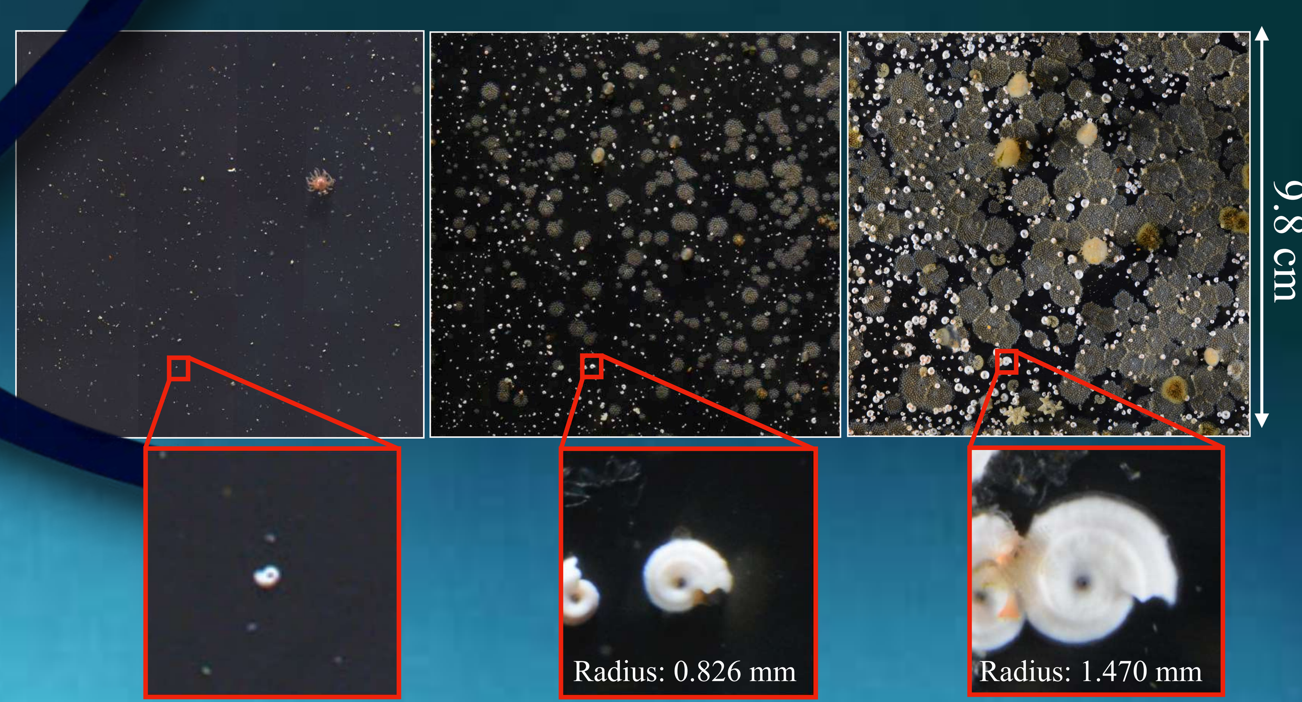
HOW DO WE HEAT THE SEA FLOOR AT THE PORTOBELLO MARINE SCIENCE LABORATORY?



We are using bespoke heated panels located on the seafloor (7 m depth) that are warmed through electrical cables and a controller on land. Panels are warmed under four treatments: ambient (control), +1°C, +2°C and heatwave (delayed warming) above ambient over extended periods (months). The responses are monitored by SCUBA diver photography. Changes will be quantified using image analysis software of plate communities.

Photographs will be merged to a single high resolution image, allowing to follow the assemblage, growth rate and competition of species through time.

Quadrat sampling of the images will be used to analyse the community composition of species through months, while image analysis software will be used to measure the individuals per species and per panel through time.



Evolution of one panel and one tube-worm through time in the Antarctic Peninsula

REFERENCES

- Ashton, G.V., Morley, S.A., Barnes, D.K.A., Clark, M.S. & Peck, L.S. (2017). Warming by 1°C Drives Species and Assemblage Level Responses in Antarctica's Marine Shallows. *Current Biology, 27*, 2698-2705.
- Clark, M.S., Nieva, L.V., Joseph I. Hoffman, J.I., Davies, A.J., Trivedi, U.H., Turner, F., Ashton. G.V. & Peck, L.S. (2019). Lack of long-term acclimatation in Antarctic encrusting species suggests vulnerability to warming. *Nature communications, 3383*(10).

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